

Linear Algebra Notes

Aleksis Arendt

Westfälische Wilhelms-Universität Münster *

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*Faculty of Mathematics and Computer Science; e-mail: aleksisdg@gmail.com

1. Jordan Forms

Bài 1.1.

Let

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ -1 & 0 & 1 \end{pmatrix}.$$

Find the minimal polynomial of A and analyze $\mathbb{R}^3$ into a direct sum of eigenspaces.

Solution. One has

$$\begin{aligned} P_A(\lambda) &= \det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 0 & 0 \\ 1 & 2-\lambda & 1 \\ -1 & 0 & -1-\lambda \end{pmatrix} \\ &= (1-\lambda) \det \begin{pmatrix} 2-\lambda & 1 \\ 0 & -1-\lambda \end{pmatrix} \\ &= (1-\lambda)(2-\lambda)(-1-\lambda) \end{aligned}$$

Since each root factor of P has degree 1, then $m_A(\lambda) = P_A(\lambda)$. Choose

$$\begin{aligned} v_1 &= (2, -1, -1) \in \ker(A - I) \\ v_2 &= (0, 1, 0) \in \ker(A - 2I) \\ v_3 &= (0, -1, 3) \in \ker(A + I), \end{aligned}$$

and let $P = (v_1, v_2, v_3)$. One can check that $\det P \neq 0$, so (v_1, v_2, v_3) forms a basis for $\mathbb{R}^3$. Hence

$$\mathbb{R}^3 = \bigoplus_{i=1}^3 \ker(A - \lambda_i I).$$

□

Bài 1.2.

Find the Jordan (or diagonal) form of

$$A = \begin{pmatrix} 3 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & -1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 2 & -2 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 1 & 0 \\ -4 & -1 & 0 \\ 4 & -8 & -2 \end{pmatrix}$$

Solution. 1. We have

$$m_A(\lambda) = P_A(\lambda) = \det \begin{pmatrix} 3-\lambda & -1 & 1 \\ 2 & -\lambda & 1 \\ 1 & -1 & 2-\lambda \end{pmatrix} = -(1-\lambda)(2-\lambda)^2,$$

Choose

$$\begin{aligned}v_1 &= (0, 1, 1) \in \ker(A - I) \\v_3 &= (1, 1, 1) \in \ker(A - 2I)^2 \setminus \ker(A - 2I) \\v_2 &= (1, 1, 0) = (A - 2I)v_3 \in \ker(A - 2I).\end{aligned}$$

We obtain

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 3 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

2. We have

$$P_B(\lambda) = \det \begin{pmatrix} 3 - \lambda & 2 & -2 \\ -1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{pmatrix} = (1 - \lambda)^3,$$

and

$$m_B(\lambda) = (\lambda - 1)^2.$$

Choose

$$\begin{aligned}v_1 &= (1, 0, 1) \in \ker(B - I) \\v_3 &= (1, 0, 0) \in \ker(B - I)^2 \setminus \ker(B - I) \\v_2 &= (2, -1, 1) = (B - I)v_3 \in \ker(B - I).\end{aligned}$$

So

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 3 & 2 & -2 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

3. We have

$$m_C(\lambda) = P_C(\lambda) = \det \begin{pmatrix} 3 - \lambda & 1 & 0 \\ -4 & -1 - \lambda & 0 \\ 4 & -8 & -2 - \lambda \end{pmatrix} = (-2 - \lambda)(1 - \lambda)^2.$$

Choose

$$\begin{aligned}v_1 &= (0, 0, 1) \in \ker(A + 2I) \\v_3 &= (-11, 7, 0) \in \ker(A - I)^2 \setminus \ker(A - I) \\v_2 &= (-15, 30, -100) = (A - I)v_3 \in \ker(A - I).\end{aligned}$$

We obtain

$$\begin{pmatrix} 0 & -15 & -11 \\ 0 & 30 & 7 \\ 1 & -100 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 3 & 1 & 0 \\ -4 & -1 & 0 \\ 4 & -8 & -2 \end{pmatrix} \begin{pmatrix} 0 & -15 & -11 \\ 0 & 30 & 7 \\ 1 & -100 & 0 \end{pmatrix} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

□

Bài 1.3.

Find a 3×3 matrix A such that $m_A(\lambda) = \lambda^2$.

Solution. Try $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$

□

2. Matrix Theory

2.1. Spectral Theorem for Matrix

We prove some version of spectral theorem, firstly the generalized version in finite Hermitian vector space, i.e a complex vector space with a positive definite inner product. Recall some definitions.

Definition 2.1 (Adjoint operator).

Let V be a finite Hermitian vector space. Let $T : V \rightarrow V$ be a linear operator, we define the adjoint T^* to be the operator satisfying

$$\langle v, T^*w \rangle = \langle Tv, w \rangle.$$

An important property about such operator is that T^* is linear, unique and if T is a matrix, then T^* is the conjugate transpose matrix of T . We call T *normal* if $TT^* = T^*T$.

Theorem 2.2 (First Spectral Theorem).

Let $T : V \rightarrow V$ be a normal linear operator. Then T admits orthonormal eigenbasis for V . Consequently, if β is any orthonormal basis for V and $A = [T]_\beta$. Then there exists $P \in U(n)$ and diagonal Σ such that $A = P\Sigma P^{-1}$.

Proof. We prove this theorem by induction by the dimension n of V . Indeed, let β be an orthonormal basis for V and let $A = [T]_\beta$.

In $\mathbb{C}$, the characteristic polynomial $\mathcal{X}_A(\lambda) = \det(A - \lambda I_n)$ always has roots. Then one can find an eigenvector $u \in V$ such that $Tu = ku$ for $k \in \mathbb{C}$.

Normalizing $\|u\| = 1$ and defining $U = \text{span}(u)$, one has $V = U \oplus U^\perp$. We will show that T can be restricted to $T|_{U^\perp} : U^\perp \rightarrow U^\perp$.

Claim 2.3.

For any subspace $W \subseteq V$ such that $T(W) \subseteq W$. Then $T^*(W^\perp) \subseteq W^\perp$.

Proof of the claim. Let $v \in W^\perp$, then for all $w \in W$, we have $Tw \in W$, hence

$$\langle w, T^*v \rangle = \langle Tw, v \rangle = 0.$$

□

Claim 2.4.

If $Tw = \lambda w$, then $T^*w = \bar{\lambda}w$.

Proof of the claim. Let $S = T - \lambda \text{Id}_V$, then one has $S^* = T^* - \bar{\lambda} \text{Id}_V$ and

$$\begin{aligned} SS^* &= (T - \lambda \text{Id}_V)(T^* - \bar{\lambda} \text{Id}_V) \\ &= TT^* - \lambda T^* - \bar{\lambda} T + |\lambda|^2 \text{Id}_V \\ &= T^*T - \lambda T^* - \bar{\lambda} T + |\lambda|^2 \text{Id}_V \\ &= (T^* - \bar{\lambda} \text{Id}_V)(T - \lambda \text{Id}_V) \\ &= S^*S \end{aligned}$$

Hence S is normal. Since $Sw = 0$, we have

$$\langle Sw, Sw \rangle = \langle w, S^*Sw \rangle = \langle w, SS^*w \rangle = \langle S^*w, S^*w \rangle = 0$$

Therefore, $S^*w = 0$. □

Claim 2.5.

One has $T(U^\perp) \subseteq U^\perp$.

Proof of the claim. Let $v \in U^\perp$, since $U = \text{span}(u)$, then $T^*u = \bar{k}u \in U$. Therefore

$$\langle Tv, u \rangle = \langle v, T^*u \rangle = 0.$$

□

Since $T|_{U^\perp} : U^\perp \rightarrow U^\perp$ and $T^*|_{U^\perp} : U^\perp \rightarrow U^\perp$, then $T|_{U^\perp}$ is also normal. By the induction hypothesis, there exists an orthonormal eigenbasis $(u_1, \dots, u_{n-1})$ for $U^\perp$. Let $u_n = u$, since $V = U \oplus U^\perp$, $(u_1, \dots, u_{n-1}, u_n)$ forms an orthonormal eigenbasis for V . Finally, let $P = [u_1, \dots, u_n]_\beta$, we will show that $P \in U(n)$. Indeed, one has

$$[P^*P]_{ij} = [\bar{P}^T P]_{ij} = \langle u_i, u_j \rangle = \delta_{ij}$$

Hence $P^*P = I_n$, this completes the proof. □

Theorem 2.6 (Second Spectral Theorem).

If $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a symmetric matrix, i.e $A^T = A$, then there exists $P \in O(n)$ and a diagonal matrix $\Sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that

$$A = P\Sigma P^T.$$

Proof. From the first theorem, it only suffices to prove some following claims.

Claim 2.7.

Every eigenvalue of A is real.

Proof of the claim. Suppose $Av = \lambda v$ with $v \neq 0$, one has

$$\lambda \langle v, v \rangle = \langle v, \lambda v \rangle = \langle v, Av \rangle = \langle v, A^*v \rangle = \langle Av, v \rangle = \bar{\lambda} \langle v, v \rangle$$

Since $\langle v, v \rangle > 0$, we obtain $\lambda = \bar{\lambda}$. □

Claim 2.8.

For each eigenvalue λ of A , one can choose a corresponding real eigenvector $Av = \lambda v$.

Proof of the claim. Let $x + iy \in \mathbb{C}^n$ satisfies $A(x + iy) = \lambda(x + iy)$, then

$$Ax + iAy = \lambda x + i\lambda y$$

Choose $v = x$, we obtain that v is real and $Av = \lambda v$. □

□

3. Inner Product Spaces

3.1. Bilinear Forms

Under coordinate changes, the area and angle between two vectors are preserved, however, we cannot describe them through linear transformations or determinants explicitly. Henceforth, we introduce the bilinear form, a concept that generalizes the inner product, and study how those changes affect this form.

Definition 3.1.

Let V be a vector space, a *bilinear form* is a map $B : V \times V \rightarrow \mathbb{R}$ that is multilinear, i.e,

$$\begin{aligned} B(\alpha u_1 + u_2, \beta v_1 + v_2) &= \alpha B(u_1, \beta v_1 + v_2) + B(u_2, \beta v_1 + v_2) \\ &= \beta B(\alpha u_1 + u_2, v_1) + B(\alpha u_1, u_2, v_2) \end{aligned}$$

for all $u_1, u_2, v_1, v_2 \in V$ and $\alpha, \beta \in \mathbb{R}$. We define the *quadratic form* of B to be $Q_B(u) = B(u, u)$ for all $u \in V$.

As we know, every linear map in a finite vector space can be represented by a matrix (or a basis). We introduce the Gram matrix, a formal way to represent any bilinear form.

Definition 3.2.

Let B be a bilinear in V and suppose $\beta = (w_1, \dots, w_n)$ is a basis of V . We define the *Gram matrix* of B in basis β to be

$$[B]_\beta = [B(w_i, w_j)]_{1 \leq i, j \leq n}$$

In particular, we define the *Gram matrix* of Q_B in basis of β to be

$$[Q_B]_\beta = [Q_B]_{ij} = \begin{cases} \left[\frac{1}{2} Q_B(u_i + u_j) - Q_B(u_i) - Q_B(u_j) \right]_{ij}, & i \neq j \\ [Q_B(u_i)]_{ii}, & i = j \end{cases}$$

If Q_B is a quadratic form of B and for any $u = (u_i)_{\beta_0}, v = (v_i)_{\beta_0}$, we write

$$B(u, v) = \sum_{i,j}^n a_{ij} u_i v_j, \quad (1)$$

and

$$Q_B(u) = \sum_{i=1}^n a_{ii} u_i^2 + \sum_{i \neq j}^n (a_{ij} + a_{ji}) u_i u_j. \quad (2)$$

Proposition 3.3.

Bilinear form is invariant under the changes of basis, i.e, one has

$$B(u, v) = [u]_{\beta}^T [B]_{\beta} [v]_{\beta}$$

for any basis β .

Lemma 3.4.

Let B be a bilinear form and suppose β_1 and β_2 are basis of V , then

$$[B]_{\beta_2} = [\beta_1 \rightarrow \beta_2]^* [B]_{\beta_1} [\beta_1 \rightarrow \beta_2],$$

and consequently,

$$[Q]_{\beta_2} = [\beta_1 \rightarrow \beta_2]^* [Q]_{\beta_1} [\beta_1 \rightarrow \beta_2].$$

Proof. Let $u, v \in V$ be arbitrary. Then we have

$$[u]_{\beta_2} = [\beta_1 \rightarrow \beta_2][u]_{\beta_1} \quad \text{and} \quad [v]_{\beta_2} = [\beta_1 \rightarrow \beta_2][v]_{\beta_1}$$

From the definition of bilinear forms

$$B(u, v) = [u]_{\beta_1}^T [B]_{\beta_1} [v]_{\beta_1} = [u]_{\beta_2}^T [B]_{\beta_2} [v]_{\beta_2}$$

In particular,

$$[u]_{\beta_1}^T [B]_{\beta_1} [v]_{\beta_1} = [u]_{\beta_2}^T [\beta_1 \rightarrow \beta_2]^T [B]_{\beta_1} [\beta_1 \rightarrow \beta_2] [v]_{\beta_2}$$

Hence,

$$[u]_{\beta_2}^T [\beta_1 \rightarrow \beta_2]^T [B]_{\beta_1} [\beta_1 \rightarrow \beta_2] [v]_{\beta_2} = [u]_{\beta_2}^T [B]_{\beta_2} [v]_{\beta_2}$$

Since this is true for all $u, v \in V$, this proves the lemma. $\square$

3.2. Lagrange's Algorithm

In this section, we establish an algorithm that transforms any bilinear form into a full-square form, which is defined as follows.

Definition 3.5.

Let B be a bilinear form, the quadratic form Q_B is called *Q-canonical* if there exists a basis β such that for any $[u]_\beta = (u_1, \dots, u_n)$, one has

$$Q_B(u) = \sum_{i=1}^n a_{ii} u_i^2, \quad \text{for all } u \in V.$$

The such basis β is called a *Q-canonical basis* for Q_B .

Theorem 3.6 (Lagrange's Algorithm).

Every quadratic form Q is *Q-canonical*.

Proof. If $Q = 0$ for all $u \in V$, then Q is *Q-canonical*. Suppose that $Q \neq 0$, we prove the theorem by induction. Consider two cases:

Case 1: There is $1 \leq i_0 \leq n$ such that $a_{i_0 i_0} \neq 0$. Let $[u]_{\beta_0} = (u_1, \dots, u_n)$, we write

$$\begin{aligned} Q(u) &= \sum_{i=1}^n a_{ii} u_i^2 + \sum_{i \neq j}^n a_{ij} u_i u_j \\ &= a_{i_0 i_0} \left[u_{i_0}^2 + 2u_{i_0} \left(\sum_{i \neq i_0} \frac{a_{i i_0} + a_{i_0 i}}{a_{i_0 i_0}} u_i \right) + \sum_{i \neq i_0} \left(\frac{a_{i i_0} + a_{i_0 i}}{a_{i_0 i_0}} u_i \right)^2 \right] \\ &\quad - a_{i_0 i_0} \sum_{i \neq i_0} \left(\frac{a_{i i_0} + a_{i_0 i}}{a_{i_0 i_0}} u_i \right)^2 + \sum_{i \neq j \neq i_0} a_{ij} u_i u_j \\ &= a_{i_0 i_0} \left(u_{i_0} + \sum_{i \neq i_0} \frac{a_{i i_0} + a_{i_0 i}}{a_{i_0 i_0}} u_i \right)^2 + \sum_{i, j \neq i_0} b_{ij} y_i y_j \\ &= a_{i_0 i_0} y_{i_0}^2 + Q_1(u) \end{aligned}$$

where Q_1 is another quadratic form which excluded u_{i_0} term, and

$$\begin{cases} y_1^1 = u_{i_0} + \sum_{i \neq i_0} \frac{a_{i i_0} + a_{i_0 i}}{a_{i_0 i_0}} u_i, \\ y_i^1 = u_i, \forall i \geq 2. \end{cases}$$

The transformation matrix of these coordinate changes is invertable.

Case 2: $a_{ii} = 0$ for all $1 \leq i \leq n$. Then there is $1 \leq i_0, j_0 \leq n$ such that $a_{i_0 j_0} \neq 0$. Let

$$\begin{cases} u_1 = y_{i_0}^1 - y_{j_0}^1 \\ u_2 = y_{i_0}^1 + y_{j_0}^1 \\ y_i^1 = u_i, \forall i \geq 3. \end{cases}$$

The transformation matrix of these coordinate changes is also invertable.

$$Q(u) = a_{i_0 j_0} (y_1^2 - y_2^2) + \sum_{i \neq j, (i, j) \neq (i_0, j_0)} a_{ij} u_i u_j$$

Since $a_{i_0j_0} \neq 0$, we deduce to the first case.

As Q is an arbitrary bilinear form, we argue similarly to Q_1 , and yield $Q_n = 0$ after n steps. Thus we obtain

$$Q(u) = \sum_{i=1}^n a_{ii}(y_i^n)^2.$$

and the n -vector set β that changes (u_i) into (y_i) defined above is a basis for V . $\square$

Since the Gram matrix of a quadratic form is symmetric, then one can relate to the *Spectral Theorem*.

Proposition 3.7.

Let Q be a quadratic form on V . Then there exists a Q -canonical basis for Q that is orthonormal.

Proof. Let β_0 be any orthonormal basis for V , then the Gram matrix $[Q]_{\beta_0}$ is symmetric, the second spectral theorem guarantees that there is $P \in O(n)$ and a scaling matrix $\Sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that

$$\Sigma = P^T[Q]_{\beta_0}P$$

Let β be the basis satisfying $P = [\beta \rightarrow \beta_0]$. Then

$$[Q]_{\beta} = [\beta \rightarrow \beta_0]^T[Q]_{\beta_0}[\beta \rightarrow \beta_0] = P^T[Q]_{\beta_0}P = \Sigma.$$

Hence β is Q -canonical and we claim that β is orthonormal. Indeed, let $u_i, u_j \in \beta$ be basis vectors. Note that since P is isometry,

$$\langle u_i, u_j \rangle = \langle Pu_i, Pu_j \rangle$$

Writing $u_i = a_{1i}v_1 + \dots + a_{ni}v_n$ and $u_j = a_{1j}v_1 + \dots + a_{nj}v_n$. Then

$$Pu_i = (a_{i1}, \dots, a_{in})$$

$$Pu_j = (a_{j1}, \dots, a_{jn}),$$

with

$$P = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \cdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{nn} \end{pmatrix}$$

Since $PP^T = I_n$, we obtain $\langle Pu_i, Pu_j \rangle = \delta_{ij}$. Hence $\langle u_i, u_j \rangle = \delta_{ij}$, which implies that β is orthonormal. $\square$

4. Euclidean Product

Definition 4.1.

Let V be a finite vector space on $\mathbb{R}$. The mapping $\langle, \rangle : V \times V \rightarrow \mathbb{R}$ is called an *inner product* if it is a symmetric and positive definite bilinear form.

Especially, we call $\langle, \rangle$ *Euclidean product* if for all $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n) \in \mathbb{R}^n$, one has

$$\langle x, y \rangle = x_1y_1 + \dots + x_ny_n.$$

5. Lagrange Multiplier Method